1) a)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 3\frac{\mathrm{d}y}{\mathrm{d}x} - 4y = 0$

- 2nd order
- Linear
- Homogeneous
- An auxiliary equation can be formed

1) b)

$y'' + y^2 = 0$

- 2nd order
- Not Linear (it contains y^2)
- Homogeneous
- An auxiliary equation cannot be formed

1) c)

$3y'' - 5y' = 7$

- 2nd order
- Linear
- Not homogeneous (RHS $\neq 0$)
- An auxiliary equation cannot be formed (auxiliary equations can only be formed from homogeneous equations)

1) d)

$y''' + 2y' - y = 0$

- 3rd order
- Linear
- Homogeneous
- An auxiliary equation can be formed (although it would be a cubic and beyond the scope of this course)

1) e)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 0$

This is:

- 2nd order
- Linear
- Homogeneous
- An auxiliary equation cannot be formed (the coefficients are not constant, as x is the coefficient of $\frac{\mathrm{d}y}{\mathrm{d}x}$)

1) f)

$\ddot{x} + 4\dot{x} + 4x = \mathrm{e}^x$

- 2nd order (note that $\ddot{x} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ and $\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t}$)
- Linear
- Not homogeneous (RHS $\neq 0$)
- An auxiliary equation cannot be formed (auxiliary equations can only be formed from homogeneous equations)

1) g)

$(y')^2 + y = 0$

- 1st order
- Not linear (contains $(y')^2$)
- Homogeneous
- An auxiliary equation cannot be formed (the equation is not linear)

1) h)

$2\ddot{x} - 3\dot{x} + x = 0$

- 2nd order (note that $\ddot{x} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ and $\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t}$)
- Linear
- Homogeneous
- An auxiliary equation can be formed

2) a)

$y = \mathrm{e}^{2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\mathrm{e}^{2x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4\mathrm{e}^{2x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 7\frac{\mathrm{d}y}{\mathrm{d}x} + 10y = 0$

LHS: $4\mathrm{e}^{2x} - 7 \times 2\mathrm{e}^{2x} + 10 \times \mathrm{e}^{2x}$

$= 4\mathrm{e}^{2x} - 14\mathrm{e}^{2x} + 10\mathrm{e}^{2x}$

$= (4 - 14 + 10)\mathrm{e}^{2x}$

$= 0$:RHS

2) b)

$y = 4\mathrm{e}^{2x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 8\mathrm{e}^{2x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 16\mathrm{e}^{2x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 7\frac{\mathrm{d}y}{\mathrm{d}x} + 10y = 0$

LHS: $16\mathrm{e}^{2x} - 7 \times 8\mathrm{e}^{2x} + 10 \times 4\mathrm{e}^{2x}$

$= 16\mathrm{e}^{2x} - 56\mathrm{e}^{2x} + 40\mathrm{e}^{2x}$

$= (16 - 56 + 40)\mathrm{e}^{2x}$

$= 0$:RHS

2) c)

$y = \mathrm{e}^{5x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 5\mathrm{e}^{5x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 25\mathrm{e}^{5x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 7\frac{\mathrm{d}y}{\mathrm{d}x} + 10y = 0$

LHS: $25\mathrm{e}^{5x} - 7 \times 5\mathrm{e}^{5x} + 10 \times \mathrm{e}^{5x}$

$= 25\mathrm{e}^{2x} - 35\mathrm{e}^{2x} + 10\mathrm{e}^{2x}$

$= (25 - 35 + 10)\mathrm{e}^{2x}$

$= 0$:RHS

2) d)

$y = 7e^{5x}$

$\frac{dy}{dx} = 35e^{5x}$

$\frac{d^2y}{dx^2} = 175e^{5x}$

$\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = 0$

LHS: $175e^{5x} - 7 \times 35e^{5x} + 10 \times 7e^{5x}$

$= 175e^{5x} - 245e^{2x} + 70e^{2x}$

$= (175 - 245 + 70)e^{2x}$

$= 0$:RHS

2) e)

$y = e^{2x} + e^{5x}$

$\frac{dy}{dx} = 2e^{2x} + 5e^{5x}$

$\frac{d^2y}{dx^2} = 4e^{2x} + 25e^{5x}$

$\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = 0$

LHS: $(4e^{2x} + 25e^{5x}) - 7(2e^{2x} + 5e^{5x}) + 10(e^{2x} + e^{5x})$

$= 4e^{2x} + 25e^{5x} - 14e^{2x} - 35e^{5x} + 10e^{2x} + 10e^{5x}$

$= (4 - 14 + 10)e^{2x} + (25 - 35 + 10)e^{5x}$

$= 0$:RHS

2) f)

$y = 4e^{2x} + 7e^{5x}$

$\frac{dy}{dx} = 8e^{2x} + 35e^{5x}$

$\frac{d^2y}{dx^2} = 16e^{2x} + 175e^{5x}$

$\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = 0$

LHS: $(16e^{2x} + 175e^{5x}) - 7(8e^{2x} + 35e^{5x}) + 10(4e^{2x} + 7e^{5x})$

$= 16e^{2x} + 175e^{5x} - 56e^{2x} - 245e^{5x} + 40e^{2x} + 70e^{5x}$

$= (16 - 56 + 40)e^{2x} + (175 - 245 + 70)e^{5x}$

$= 0$:RHS

3) a)

$y = \mathrm{e}^{mx}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 7\frac{\mathrm{d}y}{\mathrm{d}x} + 10y = 0$

$\therefore m^2\mathrm{e}^{mx} - 7m\mathrm{e}^{mx} + 10\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 7m + 10)\mathrm{e}^{mx} = 0$

$\Rightarrow m^2 - 7m + 10 = 0$ as $e^{mx} \neq 0$ for all values of m

3) b)

$m^2 - 7m + 10 = 0$

$\Rightarrow (m - 2)(m - 5) = 0$

$\Rightarrow m = 2$ or $m = 5$

Let $m_1 = 2$ and $m_2 = 5$

$y = A\mathrm{e}^{2x} + B\mathrm{e}^{5x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2A\mathrm{e}^{2x} + 5B\mathrm{e}^{5x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4A\mathrm{e}^{2x} + 25B\mathrm{e}^{5x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 7\frac{\mathrm{d}y}{\mathrm{d}x} + 10y = 0$

LHS: $(4A\mathrm{e}^{2x} + 25B\mathrm{e}^{5x}) - 7(2A\mathrm{e}^{2x} + 5B\mathrm{e}^{5x}) + 10(A\mathrm{e}^{2x} + B\mathrm{e}^{5x})$

$= 4A\mathrm{e}^{2x} + 25B\mathrm{e}^{5x} - 14A\mathrm{e}^{2x} - 35B\mathrm{e}^{5x} + 10A\mathrm{e}^{2x} + 10B\mathrm{e}^{5x}$

$= (4A - 14A + 10A)\mathrm{e}^{2x} + (25B - 35B + 10B)\mathrm{e}^{5x}$

$= 0$:RHS

4) a)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 3\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} - 3m\mathrm{e}^{mx} + 2\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 3m + 2)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 - 3m + 2 = 0$

$\Rightarrow (m-1)(m-2) = 0$

$\Rightarrow m = 1$ or $m = 2$

General Solution:

$y = A\mathrm{e}^{x} + B\mathrm{e}^{2x}$

4) b)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} - 35y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} - 2m\mathrm{e}^{mx} - 35\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 2m - 35)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 - 2m - 35 = 0$

$\Rightarrow (m-7)(m+5) = 0$

$\Rightarrow m = 7$ or $m = -5$

General Solution:

$y = A\mathrm{e}^{7x} + B\mathrm{e}^{-5x}$

4) c)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 18\frac{\mathrm{d}y}{\mathrm{d}x} + 80y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} + 18m\mathrm{e}^{mx} + 80\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 + 18m + 80)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 + 18m + 80 = 0$

$\Rightarrow (m + 10)(m + 8) = 0$

$\Rightarrow m = -10$ or $m = -8$

General Solution:

$y = A\mathrm{e}^{-10x} + B\mathrm{e}^{-8x}$

4) d)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 11\frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 0$

$\Rightarrow 2m^2\mathrm{e}^{mx} + 11m\mathrm{e}^{mx} - 6\mathrm{e}^{mx} = 0$

$\Rightarrow (2m^2 + 11m - 6)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$2m^2 + 11m - 6 = 0$

$\Rightarrow (2m - 1)(m + 6) = 0$

$\Rightarrow m = \frac{1}{2}$ or $m = -6$

General Solution:

$y = A\mathrm{e}^{\frac{1}{2}x} + B\mathrm{e}^{-6x}$

4) e)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} - 4\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 4)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 - 4 = 0$

$\Rightarrow (m - 2)(m + 2) = 0$

$\Rightarrow m = 2$ or $m = -2$

General Solution:

$y = A\mathrm{e}^{2x} + B\mathrm{e}^{-2x}$

4) f)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\Rightarrow m^2\mathrm{e}^{mx} - 4m\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 4m)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 - 4m = 0$

$\Rightarrow (m - 4)m = 0$

$\Rightarrow m = 4$ or $m = 0$

General Solution:

$y = A\mathrm{e}^{4x} + B\mathrm{e}^{0}$

$\Rightarrow y = A\mathrm{e}^{4x} + B$

5) a)

$\ddot{x} + 3\dot{x} - 28x = 0$

Auxiliary Equation:

$m^2 + 3m - 28 = 0$

$\Rightarrow (m + 7)(m - 4) = 0$

$\Rightarrow m = -7$ or $m = 4$

General Solution:

$x = Ae^{-7t} + Be^{4t}$

5) b)

$4\ddot{x} + 4\dot{x} - 15x = 0$

Auxiliary Equation:

$4m^2 + 4m - 15 = 0$

$\Rightarrow (2m + 5)(2m - 3) = 0$

$\Rightarrow m = -\frac{5}{2}$ or $m = \frac{3}{2}$

General Solution:

$x = Ae^{-\frac{5}{2}t} + Be^{\frac{3}{2}t}$

6)

$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0$

Auxiliary Equation:

$m^2 + m - 2 = 0$

$\Rightarrow (m+2)(m-1) = 0$

$\Rightarrow m = -2$ or $m = 1$

General Solution:

$y = Ae^{-2x} + Be^{x}$

Substitute $y = 2$ and $x = 0$:

$2 = Ae^{0} + Be^{0}$

$\Rightarrow A + B = 2$ (1)

Substitute $y = 3$ and $x = \ln 2$:

$3 = Ae^{-2\ln 2} + Be^{\ln 2}$

$\Rightarrow 3 = Ae^{\ln 2^{-2}} + Be^{\ln 2}$

$\Rightarrow \frac{1}{4}A + 2B = 3$ (2)

$\therefore A = \frac{4}{7}$ and $B = \frac{10}{7}$

Particular Solution:

$y = \frac{4}{7}e^{-2x} + \frac{10}{7}e^{x}$

7)

$2y'' - 7y' + 3y = 0$

Auxiliary Equation:

$2m^2 - 7m + 3 = 0$

$\Rightarrow (2m - 1)(m - 3) = 0$

$\Rightarrow m = \frac{1}{2}$ or $m = 3$

General Solution:

$y = Ae^{\frac{1}{2}x} + Be^{3x}$

Substitute $y = 5$ and $x = 0$:

$5 = Ae^0 + Be^0$

$\Rightarrow A + B = 5$ (1)

Substitute $y = -238$ and $x = \ln 4$:

$-238 = Ae^{\frac{1}{2}\ln 4} + Be^{3\ln 4}$

$\Rightarrow -238 = Ae^{\ln 4^{\frac{1}{2}}} + Be^{\ln 4^3}$

$\Rightarrow 2A + 64B = -238$ (2)

$\therefore A = 9$ and $B = -4$

Particular Solution:

$y = 9e^{\frac{1}{2}x} - 4e^{3x}$

8) a)

$y = e^{3x}$

$\frac{dy}{dx} = 3e^{3x}$

$\frac{d^2y}{dx^2} = 9e^{3x}$

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

LHS: $9e^{3x} - 6 \times 3e^{3x} + 9 \times e^{3x}$

$= 9e^{3x} - 18e^{3x} + 9e^{3x}$

$= (9 - 18 + 9)e^{3x}$

$= 0$:RHS

8) b)

$y = 2e^{3x}$

$\frac{dy}{dx} = 6e^{3x}$

$\frac{d^2y}{dx^2} = 18e^{3x}$

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

LHS: $18e^{3x} - 6 \times 6e^{3x} + 9 \times 2e^{3x}$

$= 18e^{3x} - 36e^{3x} + 18e^{3x}$

$= (18 - 36 + 18)e^{3x}$

$= 0$:RHS

8) c)

$y = xe^{3x}$

$\frac{dy}{dx} = 3xe^{3x} + e^{3x}$

$\frac{d^2y}{dx^2} = 9xe^{3x} + 3e^{3x} + 3e^{3x}$

$= 9xe^{3x} + 6e^{3x}$

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

LHS: $(9xe^{3x} + 6e^{3x}) - 6(3xe^{3x} + e^{3x}) + 9xe^{3x}$

$= 9xe^{3x} + 6e^{3x} - 18xe^{3x} - 6e^{3x} + 9xe^{3x}$

$= (9 - 18 + 9)xe^{3x} + (6 - 6)e^{3x}$

$= 0$:RHS

8) d)

$y = 5xe^{3x}$

$\frac{dy}{dx} = 15xe^{3x} + 5e^{3x}$

$\frac{d^2y}{dx^2} = 45xe^{3x} + 15e^{3x} + 15e^{3x}$

$= 45xe^{3x} + 30e^{3x}$

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

LHS: $(45xe^{3x} + 30e^{3x}) - 6(15xe^{3x} + 5e^{3x}) + 45xe^{3x}$

$= 45xe^{3x} + 30e^{3x} - 90xe^{3x} - 30e^{3x} + 45xe^{3x}$

$= (45 - 90 + 45)xe^{3x} + (30 - 30)e^{3x}$

$= 0$:RHS

8) e)

$y = \mathrm{e}^{3x} + x\mathrm{e}^{3x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3\mathrm{e}^{3x} + 3x\mathrm{e}^{3x} + \mathrm{e}^{3x}$

$= 3x\mathrm{e}^{3x} + 4\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 9x\mathrm{e}^{3x} + 3\mathrm{e}^{3x} + 12\mathrm{e}^{3x}$

$= 9x\mathrm{e}^{3x} + 15\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 6\frac{\mathrm{d}y}{\mathrm{d}x} + 9y = 0$

LHS: $(9x\mathrm{e}^{3x} + 15\mathrm{e}^{3x}) - 6(3x\mathrm{e}^{3x} + 4\mathrm{e}^{3x}) + 9(\mathrm{e}^{3x} + x\mathrm{e}^{3x})$

$= 9x\mathrm{e}^{3x} + 15\mathrm{e}^{3x} - 18x\mathrm{e}^{3x} - 24\mathrm{e}^{3x} + 9\mathrm{e}^{3x} + 9x\mathrm{e}^{3x}$

$= (9 - 18 + 9)x\mathrm{e}^{3x} + (15 - 24 + 9)\mathrm{e}^{3x}$

$= 0$:RHS

8) f)

$y = 2\mathrm{e}^{3x} + 5x\mathrm{e}^{3x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 6\mathrm{e}^{3x} + 15x\mathrm{e}^{3x} + 5\mathrm{e}^{3x}$

$= 15x\mathrm{e}^{3x} + 11\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 45x\mathrm{e}^{3x} + 15\mathrm{e}^{3x} + 33\mathrm{e}^{3x}$

$= 45x\mathrm{e}^{3x} + 48\mathrm{e}^{3x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 6\frac{\mathrm{d}y}{\mathrm{d}x} + 9y = 0$

LHS: $(45x\mathrm{e}^{3x} + 48\mathrm{e}^{3x}) - 6(15x\mathrm{e}^{3x} + 11\mathrm{e}^{3x}) + 9(2\mathrm{e}^{3x} + 5x\mathrm{e}^{3x})$

$= 45x\mathrm{e}^{3x} + 48\mathrm{e}^{3x} - 90x\mathrm{e}^{3x} - 66\mathrm{e}^{3x} + 18\mathrm{e}^{3x} + 45x\mathrm{e}^{3x}$

$= (45 - 90 + 45)x\mathrm{e}^{3x} + (48 - 66 + 18)\mathrm{e}^{3x}$

$= 0$:RHS

9) a)

$y = xe^{mx}$

$\frac{dy}{dx} = mxe^{mx} + e^{mx}$

$\frac{d^2y}{dx^2} = m^2xe^{mx} + me^{mx} + me^{mx}$

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

$\therefore (m^2xe^{mx} + me^{mx} + me^{mx}) - 6(mxe^{mx} + e^{mx}) + 9(xe^{mx}) = 0$

$\Rightarrow m^2xe^{mx} + me^{mx} + me^{mx} - 6mxe^{mx} - 6e^{mx} + 9xe^{mx} = 0$

$\Rightarrow (m^2 - 6m + 9)xe^{mx} + (2m - 6)e^{mx} = 0$

$\Rightarrow \big((m^2 - 6m + 9)x + (2m - 6)\big)e^{mx} = 0$

$\Rightarrow m^2 - 6m + 9 = 0$ and $2m - 6 = 0$ as $e^{mx} \neq 0$ for all values of m

9) b)

Firstly:

$m^2 - 6m + 9 = 0 \Rightarrow (m - 3)^2 = 0 \Rightarrow m = 3$

Secondly:

$2m - 6 = 0 \Rightarrow m = 3$

So $m = 3$ in order to satisfy both equations

$y = Axe^{3x} + Be^{3x}$

$\frac{dy}{dx} = 3Axe^{3x} + Ae^{3x} + 3Be^{3x}$

$\frac{d^2y}{dx^2} = 9Axe^{3x} + 3Ae^{3x} + 3Ae^{3x} + 9Be^{3x}$

$\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

LHS: $(9Axe^{3x} + 3Ae^{3x} + 3Ae^{3x} + 9Be^{3x}) - 6(3Axe^{3x} + Ae^{3x} + 3Be^{3x}) +$
$9(Axe^{3x} + Be^{3x})$

$= 9Axe^{3x} + 3Ae^{3x} + 3Ae^{3x} + 9Be^{3x} - 18Axe^{3x} - 6Ae^{3x} - 18Be^{3x} + 9Axe^{3x} + 9Be^{3x}$

$= (9A - 18A + 9A)xe^{3x} + (3A + 3A + 9B - 6A - 18B + 9B)e^{3x}$

$= 0$:RHS

10) a)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 10\frac{\mathrm{d}y}{\mathrm{d}x} + 25y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} - 10m\mathrm{e}^{mx} + 25\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 10m + 25)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 - 10m + 25 = 0$

$\Rightarrow (m-5)^2 = 0$

$\Rightarrow m = 5$

General Solution:

$y = Ax\mathrm{e}^{5x} + B\mathrm{e}^{5x}$

10) b)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 12\frac{\mathrm{d}y}{\mathrm{d}x} + 36y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} + 12m\mathrm{e}^{mx} + 36\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 + 12m + 36)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 + 12m + 36 = 0$

$\Rightarrow (m+6)^2 = 0$

$\Rightarrow m = -6$

General Solution:

$y = Ax\mathrm{e}^{-6x} + B\mathrm{e}^{-6x}$

10) c)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 12\frac{\mathrm{d}y}{\mathrm{d}x} + 9y = 0$

$\Rightarrow 4m^2\mathrm{e}^{mx} + 12m\mathrm{e}^{mx} + 9\mathrm{e}^{mx} = 0$

$\Rightarrow (4m^2 + 12m + 9)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$4m^2 + 12m + 9 = 0$

$\Rightarrow (2m + 3)^2 = 0$

$\Rightarrow m = -\frac{3}{2}$

General Solution:

$y = Ax\mathrm{e}^{-\frac{3}{2}x} + B\mathrm{e}^{-\frac{3}{2}x}$

10) d)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 42\frac{\mathrm{d}y}{\mathrm{d}x} + 49y = 0$

$\Rightarrow 9m^2\mathrm{e}^{mx} - 42m\mathrm{e}^{mx} + 49\mathrm{e}^{mx} = 0$

$\Rightarrow (9m^2 - 42m + 49)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$9m^2 - 42m + 49 = 0$

$\Rightarrow (3m - 7)^2 = 0$

$\Rightarrow m = \frac{7}{3}$

General Solution:

$y = Ax\mathrm{e}^{\frac{7}{3}x} + B\mathrm{e}^{\frac{7}{3}x}$

11) a)

$y'' + 26y' + 169y = 0$

Auxiliary Equation:

$m^2 + 26m + 169 = 0$

$\Rightarrow (m + 13)^2 = 0$

$\Rightarrow m = -13$

General Solution:

$y = Axe^{-13x} + Be^{-13x}$

11) b)

$9y'' - 48y' + 64y = 0$

Auxiliary Equation:

$9m^2 - 48m + 64 = 0$

$\Rightarrow (3m - 8)^2 = 0$

$\Rightarrow m = \frac{8}{3}$

General Solution:

$y = Axe^{\frac{8}{3}x} + Be^{\frac{8}{3}x}$

12)

$\ddot{x} - 10\dot{x} + 25x = 0$

Auxiliary Equation:

$m^2 - 10m + 25 = 0$

$\Rightarrow (m-5)^2 = 0$

$\Rightarrow m = 5$

General Solution:

$x = Ate^{5t} + Be^{5t}$

Substitute $x = 0$ and $t = 0$:

$0 = A \times 0 \times e^0 + Be^0$

$\Rightarrow B = 0$

Substitute $x = 4$ and $t = 1$:

$4 = A \times 1 \times e^{5\times 1} + 0$

$\Rightarrow 4 = Ae^5$

$\Rightarrow A = 4e^{-5}$

Particular Solution:

$y = 4e^{-5}te^{5t}$

$\Rightarrow y = 4te^{5t-5}$

13)

$9\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 6\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0$

Auxiliary Equation:

$9m^2 - 6m + 1 = 0$

$\Rightarrow (3m-1)^2 = 0$

$\Rightarrow m = \frac{1}{3}$

General Solution:

$y = Ax\mathrm{e}^{\frac{1}{3}x} + B\mathrm{e}^{\frac{1}{3}x}$

Substitute $y = 3$ and $x = 3$:

$3 = A \times 3 \times \mathrm{e}^{\frac{1}{3}\times 3} + B\mathrm{e}^{\frac{1}{3}\times 3}$

$\Rightarrow 3 = 3A\mathrm{e} + B\mathrm{e}$

$\Rightarrow 3A + B = \frac{3}{\mathrm{e}}$ (1)

Substitute $y = 4$ and $x = 6$:

$4 = A \times 6 \times \mathrm{e}^{\frac{1}{3}\times 6} + B\mathrm{e}^{\frac{1}{3}\times 6}$

$\Rightarrow 4 = 6A\mathrm{e}^2 + B\mathrm{e}^2$

$\Rightarrow 6A + B = \frac{4}{\mathrm{e}^2}$ (2)

(2) – (1):

$3A = \frac{4}{\mathrm{e}^2} - \frac{3}{\mathrm{e}}$

$\Rightarrow A = \frac{4}{3\mathrm{e}^2} - \frac{1}{\mathrm{e}} = \frac{4-3\mathrm{e}}{3\mathrm{e}^2}$

Substitute into (1):

$3 \times \frac{4-3\mathrm{e}}{3\mathrm{e}^2} + B = \frac{3}{\mathrm{e}}$

$\Rightarrow \frac{4-3\mathrm{e}}{\mathrm{e}^2} + B = \frac{3}{\mathrm{e}}$

$\Rightarrow B = \frac{3}{\mathrm{e}} - \frac{4-3\mathrm{e}}{\mathrm{e}^2} = \frac{3\mathrm{e}-(4-3\mathrm{e})}{\mathrm{e}^2} = \frac{6\mathrm{e}-4}{\mathrm{e}^2}$

Particular Solution:

$y = \left(\frac{4-3\mathrm{e}}{3\mathrm{e}^2}\right)x\mathrm{e}^{\frac{1}{3}x} + \left(\frac{6\mathrm{e}-4}{\mathrm{e}^2}\right)\mathrm{e}^{\frac{1}{3}x}$

$\Rightarrow y = \frac{1}{3}(4-3\mathrm{e})x\mathrm{e}^{\frac{1}{3}x-2} + (6\mathrm{e}-4)\mathrm{e}^{\frac{1}{3}x-2}$

14) a)

$y = \mathrm{e}^{2\mathrm{i}x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\mathrm{i}\mathrm{e}^{2\mathrm{i}x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -4\mathrm{e}^{2\mathrm{i}x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$

LHS: $-4\mathrm{e}^{2\mathrm{i}x} + 4\mathrm{e}^{2\mathrm{i}x}$

$= 0$:RHS

14) b)

$y = \mathrm{e}^{-2\mathrm{i}x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -2\mathrm{i}\mathrm{e}^{-2\mathrm{i}x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -4\mathrm{e}^{-2\mathrm{i}x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$

LHS: $-4\mathrm{e}^{-2\mathrm{i}x} + 4\mathrm{e}^{-2\mathrm{i}x}$

$= 0$:RHS

14) c)

$y = 2\mathrm{e}^{2\mathrm{i}x} - 3\mathrm{e}^{-2\mathrm{i}x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4\mathrm{i}\mathrm{e}^{2\mathrm{i}x} + 6\mathrm{i}\mathrm{e}^{-2\mathrm{i}x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -8\mathrm{e}^{2\mathrm{i}x} + 12\mathrm{e}^{-2\mathrm{i}x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$

LHS: $\left(-8\mathrm{e}^{2\mathrm{i}x} + 12\mathrm{e}^{-2\mathrm{i}x}\right) + 4\left(2\mathrm{e}^{2\mathrm{i}x} - 3\mathrm{e}^{-2\mathrm{i}x}\right)$

$= -8\mathrm{e}^{2\mathrm{i}x} + 12\mathrm{e}^{-2\mathrm{i}x} + 8\mathrm{e}^{2\mathrm{i}x} - 12\mathrm{e}^{-2\mathrm{i}x}$

$= 0$:RHS

15)

$e^{2ix} = \cos 2x + i \sin 2x$

$e^{-2ix} = \cos(-2x) + i \sin(-2x)$

$= \cos 2x - i \sin 2x$

$Ae^{2ix} + Be^{-2ix}$

$= A(\cos 2x + i \sin 2x) + B(\cos 2x - i \sin 2x)$

$= (A + B) \cos 2x + (A - B)i \sin 2x$

Since we are working with differential equations that will have real coefficients, for real-valued solutions, we choose A and B as complex conjugates:

Let $A = p + qi$ and $B = p - qi$

Then let $C = A + B = p + qi + p - qi = 2p$

and let $D = (A - B)i = (p + qi - p + qi)i = -2q$

So C and D are real constants

So we have that

$Ae^{2ix} + Be^{-2ix} = C \cos 2x + D \sin 2x$

16) a)

$y = \cos 2x$

$\frac{dy}{dx} = -2 \sin 2x$

$\frac{d^2y}{dx^2} = -4 \cos 2x$

$\frac{d^2y}{dx^2} + 4y = 0$

LHS: $-4 \cos 2x + 4 \cos 2x$

$= 0$:RHS

16) b)

$y = \sin 2x$

$\frac{dy}{dx} = 2 \cos 2x$

$\frac{d^2y}{dx^2} = -4 \sin 2x$

$\frac{d^2y}{dx^2} + 4y = 0$

LHS: $-4 \sin 2x + 4 \sin 2x$

$= 0$:RHS

16) c)

$y = 3\cos 2x$

$\frac{dy}{dx} = -6\sin 2x$

$\frac{d^2y}{dx^2} = -12\cos 2x$

$\frac{d^2y}{dx^2} + 4y = 0$

LHS: $-12\cos 2x + 4 \times 3\cos 2x$

$= 0$:RHS

16) d)

$y = 5\sin 2x$

$\frac{dy}{dx} = 10\cos 2x$

$\frac{d^2y}{dx^2} = -20\sin 2x$

$\frac{d^2y}{dx^2} + 4y = 0$

LHS: $-20\sin 2x + 4 \times 5\sin 2x$

$= 0$:RHS

16) e)

$y = \cos 2x + \sin 2x$

$\frac{dy}{dx} = -2\sin 2x + 2\cos 2x$

$\frac{d^2y}{dx^2} = -4\cos 2x - 4\sin 2x$

$\frac{d^2y}{dx^2} + 4y = 0$

LHS: $(-4\cos 2x - 4\sin 2x) + 4(\cos 2x + \sin 2x)$

$= -4\cos 2x - 4\sin 2x + 4\cos 2x + 4\sin 2x$

$= 0$:RHS

16) f)

$y = 3\cos 2x + 5\sin 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -6\sin 2x + 10\cos 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -12\cos 2x - 20\sin 2x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$

LHS: $(-12\cos 2x - 20\sin 2x) + 4(3\cos 2x + 5\sin 2x)$

$= -12\cos 2x - 20\sin 2x + 12\cos 2x + 20\sin 2x$

$= 0$:RHS

17) a)

$y = \mathrm{e}^{mx}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = 0$

$\therefore m^2\mathrm{e}^{mx} + 9\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 + 9)\mathrm{e}^{mx} = 0$

$\Rightarrow m^2 + 9 = 0$ as $\mathrm{e}^{mx} \neq 0$ for all values of m

17) b)

$m^2 + 9 = 0 \Rightarrow m = \pm 3\mathrm{i}$

$y = A\cos 3x + B\sin 3x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -3A\sin 3x + 3B\cos 3x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -9A\cos 3x - 9B\sin 3x$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = 0$

LHS: $(-9A\cos 3x - 9B\sin 3x) + 9(A\cos 3x + B\sin 3x)$

$= -9A\cos 3x - 9B\sin 3x + 9A\cos 3x + 9B\sin 3x$

$= 0$:RHS

18) a)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 25y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} + 25\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 + 25)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 + 25 = 0$

$\Rightarrow m = \pm 5\mathrm{i}$

General Solution:

$y = A\cos 5x + B\sin 5x$

18) b)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 49y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} + 49\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 + 49)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 + 49 = 0$

$\Rightarrow m = \pm 7\mathrm{i}$

General Solution:

$y = A\cos 7x + B\sin 7x$

18) c)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$4\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 9y = 0$

$\Rightarrow 4m^2\mathrm{e}^{mx} + 9\mathrm{e}^{mx} = 0$

$\Rightarrow (4m^2 + 9)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$4m^2 + 9 = 0$

$\Rightarrow m = \pm\frac{3}{2}\mathrm{i}$

General Solution:

$y = A\cos\frac{3}{2}x + B\sin\frac{3}{2}x$

19) a)

$\ddot{x} + 8x = 0$

Auxiliary Equation:

$m^2 + 8 = 0$

$\Rightarrow m = \pm 2\sqrt{2}\mathrm{i}$

General Solution:

$x = A\cos\left(2\sqrt{2}t\right) + B\sin\left(2\sqrt{2}t\right)$

19) b)

$9\ddot{x} + 25x = 0$

Auxiliary Equation:

$9m^2 + 25 = 0$

$\Rightarrow m = \pm\frac{5}{3}\mathrm{i}$

General Solution:

$x = A\cos\frac{5}{3}t + B\sin\frac{5}{3}t$

20)

$y'' + 100y = 0$

Auxiliary Equation:

$m^2 + 100 = 0$

$\Rightarrow m = \pm 10\text{i}$

General Solution:

$y = A\cos 10x + B\sin 10x$

Substitute $y = 0$ and $x = 0$:

$0 = A\cos 0 + B\sin 0$

$\Rightarrow A = 0$

Substitute $y = 3$ and $x = \frac{\pi}{20}$:

$3 = B\sin\left(10 \times \frac{\pi}{20}\right)$

$\Rightarrow 3 = B\sin\frac{\pi}{2}$

$\Rightarrow B = 3$

Particular Solution:

$y = 3\sin 10x$

21)

$16\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + y = 0$

Auxiliary Equation:

$16m^2 + 1 = 0$

$\Rightarrow m = \pm\frac{1}{4}\mathrm{i}$

General Solution:

$y = A\cos\frac{1}{4}x + B\sin\frac{1}{4}x$

Substitute $y = 2$ and $x = \pi$:

$2 = A\cos\frac{\pi}{4} + B\sin\frac{\pi}{4}$

$\Rightarrow 2 = \frac{\sqrt{2}}{2}A + \frac{\sqrt{2}}{2}B$

$\Rightarrow A + B = 2\sqrt{2}$ (1)

Substitute $y = -3$ and $x = 2\pi$:

$-3 = A\cos\frac{\pi}{2} + B\sin\frac{\pi}{2}$

$\Rightarrow B = -3$

Substitute into (1):

$A - 3 = 2\sqrt{2}$

$\Rightarrow A = 3 + 2\sqrt{2}$

Particular Solution:

$y = \left(3 + 2\sqrt{2}\right)\cos\frac{1}{4}x - 3\sin\frac{1}{4}x$

22) a)

$y = \mathrm{e}^{mx}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0$

$\therefore m^2\mathrm{e}^{mx} - 4m\mathrm{e}^{mx} + 13\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 4m + 13)\mathrm{e}^{mx} = 0$

$\Rightarrow m^2 - 4m + 13 = 0$ as $\mathrm{e}^{mx} \neq 0$ for all values of m

22) b)

$m^2 - 4m + 13 = 0$

$\Rightarrow (m-2)^2 - 4 + 13 = 0$

$\Rightarrow (m-2)^2 = -9$

$\Rightarrow m = 2 \pm 3\mathrm{i}$

23) a)

$y = \mathrm{e}^{(2+3\mathrm{i})x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = (2+3\mathrm{i})\mathrm{e}^{(2+3\mathrm{i})x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = (2+3\mathrm{i})^2\mathrm{e}^{(2+3\mathrm{i})x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0$

LHS: $(2+3\mathrm{i})^2\mathrm{e}^{(2+3\mathrm{i})x} - 4(2+3\mathrm{i})\mathrm{e}^{(2+3\mathrm{i})x} + 13\mathrm{e}^{(2+3\mathrm{i})x}$

$= ((2+3\mathrm{i})^2 - 4(2+3\mathrm{i}) + 13)\mathrm{e}^{(2+3\mathrm{i})x}$

$= \big((4+12\mathrm{i}-9) - 4(2+3\mathrm{i}) + 13\big)\mathrm{e}^{(2+3\mathrm{i})x}$

$= (-5+12\mathrm{i}-8-12\mathrm{i}+13)\mathrm{e}^{(2+3\mathrm{i})x}$

$= 0$:RHS

23) b)

$y = \mathrm{e}^{(2-3\mathrm{i})x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = (2-3\mathrm{i})\mathrm{e}^{(2-3\mathrm{i})x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = (2-3\mathrm{i})^2\mathrm{e}^{(2-3\mathrm{i})x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0$

LHS: $(2-3\mathrm{i})^2\mathrm{e}^{(2-3\mathrm{i})x} - 4(2-3\mathrm{i})\mathrm{e}^{(2-3\mathrm{i})x} + 13\mathrm{e}^{(2-3\mathrm{i})x}$

$= ((2-3\mathrm{i})^2 - 4(2-3\mathrm{i}) + 13)\mathrm{e}^{(2-3\mathrm{i})x}$

$= \big((4-12\mathrm{i}-9) - 4(2-3\mathrm{i}) + 13\big)\mathrm{e}^{(2+3\mathrm{i})x}$

$= (-5-12\mathrm{i}-8+12\mathrm{i}+13)\mathrm{e}^{(2+3\mathrm{i})x}$

$= 0$:RHS

23) c)

$y = 5\mathrm{e}^{(2+3\mathrm{i})x} + 2\mathrm{e}^{(2-3\mathrm{i})x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 5(2+3\mathrm{i})\mathrm{e}^{(2+3\mathrm{i})x} + 2(2-3\mathrm{i})\mathrm{e}^{(2-3\mathrm{i})x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 5(2+3\mathrm{i})^2\mathrm{e}^{(2+3\mathrm{i})x} + 2(2-3\mathrm{i})^2\mathrm{e}^{(2-3\mathrm{i})x}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0$

LHS: $\left(5(2+3\mathrm{i})^2\mathrm{e}^{(2+3\mathrm{i})x} + 2(2-3\mathrm{i})^2\mathrm{e}^{(2-3\mathrm{i})x}\right) - 4\left(5(2+3\mathrm{i})\mathrm{e}^{(2+3\mathrm{i})x} + 2(2-3\mathrm{i})\mathrm{e}^{(2-3\mathrm{i})x}\right) + 13\left(5\mathrm{e}^{(2+3\mathrm{i})x} + 2\mathrm{e}^{(2-3\mathrm{i})x}\right)$

$= (5(2+3\mathrm{i})^2 - 20(2+3\mathrm{i}) + 65)\mathrm{e}^{(2+3\mathrm{i})x} + (2(2-3\mathrm{i})^2 - 8(2-3\mathrm{i}) + 26)\mathrm{e}^{(2-3\mathrm{i})x}$

$= (5(4+12\mathrm{i}-9) - (40+60\mathrm{i}) + 65)\mathrm{e}^{(2+3\mathrm{i})x} + (2(4-12\mathrm{i}-9) - (16-24\mathrm{i}) + 26)\mathrm{e}^{(2-3\mathrm{i})x}$

$= (-25+60\mathrm{i}-40-60\mathrm{i}+65)\mathrm{e}^{(2+3\mathrm{i})x} + (-10-24\mathrm{i}-16+24\mathrm{i}+26)\mathrm{e}^{(2-3\mathrm{i})x}$

$= 0$:RHS

24)

$\mathrm{e}^{(2+3\mathrm{i})x} = \mathrm{e}^{2x}\mathrm{e}^{3\mathrm{i}x} = \mathrm{e}^{2x}(\cos 3x + \mathrm{i}\sin 3x)$

$\mathrm{e}^{(2-3\mathrm{i})x} = \mathrm{e}^{2x}\mathrm{e}^{-3\mathrm{i}x} = \mathrm{e}^{2x}(\cos(-3x) + \mathrm{i}\sin(-3x))$

$= \mathrm{e}^{2x}(\cos 3x - \mathrm{i}\sin 3x)$

$A\mathrm{e}^{(2+3\mathrm{i})x} + B\mathrm{e}^{(2-3\mathrm{i})x}$

$= A\mathrm{e}^{2x}(\cos 3x + \mathrm{i}\sin 3x) + B\mathrm{e}^{2x}(\cos 3x - \mathrm{i}\sin 3x)$

$= (A+B)\mathrm{e}^{2x}\cos 3x + (A-B)\mathrm{i}\mathrm{e}^{2x}\sin 3x$

Since we are working with differential equations that will have real coefficients, for real-valued solutions, we choose A and B as complex conjugates:

Let $A = p + q\mathrm{i}$ and $B = p - q\mathrm{i}$

Then let $C = A + B = p + q\mathrm{i} + p - q\mathrm{i} = 2p$

and let $D = (A-B)\mathrm{i} = (p + q\mathrm{i} - p + q\mathrm{i})\mathrm{i} = -2q$

So C and D are real constants

So we have that

$A\mathrm{e}^{(2+3\mathrm{i})x} + B\mathrm{e}^{(2-3\mathrm{i})x} = C\mathrm{e}^{2x}\cos 3x + D\mathrm{e}^{2x}\sin 3x$

$\Rightarrow A\mathrm{e}^{(2+3\mathrm{i})x} + B\mathrm{e}^{(2-3\mathrm{i})x} = \mathrm{e}^{2x}(C\cos 3x + D\sin 3x)$

25) a)

$y = e^{2x}\cos 3x$

$\frac{dy}{dx} = -3e^{2x}\sin 3x + 2e^{2x}\cos 3x$

$\frac{d^2y}{dx^2} = -9e^{2x}\cos 3x - 6e^{2x}\sin 3x - 6e^{2x}\sin 3x + 4e^{2x}\cos 3x$

$= -5e^{2x}\cos 3x - 12e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$

LHS: $(-5e^{2x}\cos 3x - 12e^{2x}\sin 3x) - 4(-3e^{2x}\sin 3x + 2e^{2x}\cos 3x) + 13e^{2x}\cos 3x$

$= -5e^{2x}\cos 3x - 12e^{2x}\sin 3x + 12e^{2x}\sin 3x - 8e^{2x}\cos 3x + 13e^{2x}\cos 3x$

$= (-5 - 8 + 13)e^{2x}\cos 3x + (-12 + 12)e^{2x}\sin 3x$

$= 0$:RHS

25) b)

$y = e^{2x}\sin 3x$

$\frac{dy}{dx} = 3e^{2x}\cos 3x + 2e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} = -9e^{2x}\sin 3x + 6e^{2x}\cos 3x + 6e^{2x}\cos 3x + 4e^{2x}\sin 3x$

$= 12e^{2x}\cos 3x - 5e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$

LHS: $(12e^{2x}\cos 3x - 5e^{2x}\sin 3x) - 4(3e^{2x}\cos 3x + 2e^{2x}\sin 3x) + 13e^{2x}\sin 3x$

$= 12e^{2x}\cos 3x - 5e^{2x}\sin 3x - 12e^{2x}\cos 3x - 8e^{2x}\sin 3x + 13e^{2x}\sin 3x$

$= (12 - 12)e^{2x}\cos 3x + (-5 - 8 + 13)e^{2x}\sin 3x$

$= 0$:RHS

25) c)

$y = 4e^{2x}\cos 3x$

$\frac{dy}{dx} = -12e^{2x}\sin 3x + 8e^{2x}\cos 3x$

$\frac{d^2y}{dx^2} = -36e^{2x}\cos 3x - 24e^{2x}\sin 3x - 24e^{2x}\sin 3x + 16e^{2x}\cos 3x$

$= -20e^{2x}\cos 3x - 48e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$

LHS: $(-20e^{2x}\cos 3x - 48e^{2x}\sin 3x) - 4(-12e^{2x}\sin 3x + 8e^{2x}\cos 3x) + 52e^{2x}\cos 3x$

$= -20e^{2x}\cos 3x - 48e^{2x}\sin 3x + 48e^{2x}\sin 3x - 32e^{2x}\cos 3x + 52e^{2x}\cos 3x$

$= (-20 - 32 + 52)e^{2x}\cos 3x + (-48 + 48)e^{2x}\sin 3x$

$= 0$:RHS

25) d)

$y = 5e^{2x}\sin 3x$

$\frac{dy}{dx} = 15e^{2x}\cos 3x + 10e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} = -45e^{2x}\sin 3x + 30e^{2x}\cos 3x + 30e^{2x}\cos 3x + 20e^{2x}\sin 3x$

$= 60e^{2x}\cos 3x - 25e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$

LHS: $(60e^{2x}\cos 3x - 25e^{2x}\sin 3x) - 4(15e^{2x}\cos 3x + 10e^{2x}\sin 3x) + 65e^{2x}\sin 3x$

$= 60e^{2x}\cos 3x - 25e^{2x}\sin 3x - 60e^{2x}\cos 3x - 40e^{2x}\sin 3x + 65e^{2x}\sin 3x$

$= (60 - 60)e^{2x}\cos 3x + (-25 - 40 + 65)e^{2x}\sin 3x$

$= 0$:RHS

25) e)

$y = e^{2x}\cos 3x + e^{2x}\sin 3x$

$\frac{dy}{dx} = -3e^{2x}\sin 3x + 2e^{2x}\cos 3x + 3e^{2x}\cos 3x + 2e^{2x}\sin 3x$

$= 5e^{2x}\cos 3x - e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} = 10e^{2x}\cos 3x - 15e^{2x}\sin 3x - 2e^{2x}\sin 3x - 3e^{2x}\cos 3x$

$= 7e^{2x}\cos 3x - 17e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$

LHS: $(7e^{2x}\cos 3x - 17e^{2x}\sin 3x) - 4(5e^{2x}\cos 3x - e^{2x}\sin 3x) + 13(e^{2x}\cos 3x + e^{2x}\sin 3x)$

$= 7e^{2x}\cos 3x - 17e^{2x}\sin 3x - 20e^{2x}\cos 3x + 4e^{2x}\sin 3x + 13e^{2x}\cos 3x + 13e^{2x}\sin 3x$

$= (7 - 20 + 13)e^{2x}\cos 3x + (-17 + 4 + 13)e^{2x}\sin 3x$

$= 0$:RHS

25) f)

$y = 4e^{2x}\cos 3x + 5e^{2x}\sin 3x$

$\frac{dy}{dx} = -12e^{2x}\sin 3x + 8e^{2x}\cos 3x + 15e^{2x}\cos 3x + 10e^{2x}\sin 3x$

$23e^{2x}\cos 3x - 2e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} = 46e^{2x}\cos 3x - 69e^{2x}\sin 3x - 4e^{2x}\sin 3x - 6e^{2x}\cos 3x$

$= 40e^{2x}\cos 3x - 73e^{2x}\sin 3x$

$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$

LHS: $(40e^{2x}\cos 3x - 73e^{2x}\sin 3x) - 4(23e^{2x}\cos 3x - 2e^{2x}\sin 3x) + 13(4e^{2x}\cos 3x + 5e^{2x}\sin 3x)$

$= 40e^{2x}\cos 3x - 73e^{2x}\sin 3x - 92e^{2x}\cos 3x + 8e^{2x}\sin 3x + 52e^{2x}\cos 3x + 65e^{2x}\sin 3x$

$= (40 - 92 + 52)e^{2x}\cos 3x + (-73 + 8 + 65)e^{2x}\sin 3x$

$= 0$:RHS

26) a)

$y = \mathrm{e}^{mx}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 29y = 0$

$\therefore m^2\mathrm{e}^{mx} - 4m\mathrm{e}^{mx} + 29\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 4m + 29)\mathrm{e}^{mx} = 0$

$\Rightarrow m^2 - 4m + 29 = 0$ as $\mathrm{e}^{mx} \neq 0$ for all values of m

26) b)

$m^2 - 4m + 29 = 0$

$\Rightarrow (m-2)^2 - 4 + 29 = 0$

$\Rightarrow (m-2)^2 = -25$

$\Rightarrow m = 2 \pm 5\mathrm{i}$

26) c)

$y = \mathrm{e}^{2x}(A\cos 5x + B\sin 5x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{2x}(-5A\sin 5x + 5B\cos 5x) + 2\mathrm{e}^{2x}(A\cos 5x + B\sin 5x)$

$= \mathrm{e}^{2x}(-5A\sin 5x + 5B\cos 5x + 2A\cos 5x + 2B\sin 5x)$

$= \mathrm{e}^{2x}\big((-5A + 2B)\sin 5x + (2A + 5B)\cos 5x\big)$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \mathrm{e}^{2x}\big((-25A + 10B)\cos 5x - (10A + 25B)\sin 5x\big) + 2\mathrm{e}^{2x}\big((-5A + 2B)\sin 5x + (2A + 5B)\cos 5x\big)$

$= \mathrm{e}^{2x}\big((-25A + 10B)\cos 5x - (10A + 25B)\sin 5x\big) + \mathrm{e}^{2x}\big((-10A + 4B)\sin 5x + (4A + 10B)\cos 5x\big)$

$= \mathrm{e}^{2x}\big((-21A + 20B)\cos 5x + (-20A - 21B)\sin 5x\big)$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 29y = 0$

LHS: $\mathrm{e}^{2x}\big((-21A + 20B)\cos 5x + (-20A - 21B)\sin 5x\big) - 4\mathrm{e}^{2x}\big((-5A + 2B)\sin 5x + (2A + 5B)\cos 5x\big) + 29\mathrm{e}^{2x}(A\cos 5x + B\sin 5x)$

$= \mathrm{e}^{2x}\big((-21A + 20B)\cos 5x + (-20A - 21B)\sin 5x\big) + \mathrm{e}^{2x}\big((20A - 8B)\sin 5x + (-8A - 20B)\cos 5x\big) + \mathrm{e}^{2x}(29A\cos 5x + 29B\sin 5x)$

$= \mathrm{e}^{2x}\big((-21A + 20B - 8A - 20B + 29A)\cos 5x + (-20A - 21B + 20A - 8B + 29B)\sin 5x\big)$

$= 0$:RHS

27) a)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + 5y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} - 2m\mathrm{e}^{mx} + 5\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 2m + 5)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 - 2m + 5 = 0$

$\Rightarrow (m-1)^2 - 1 + 5 = 0$

$\Rightarrow (m-1)^2 = -4$

$\Rightarrow m = 1 \pm 2\mathrm{i}$

General Solution:

$y = \mathrm{e}^x(A\cos 2x + B\sin 2x)$

27) b)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 6\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} + 6m\mathrm{e}^{mx} + 13\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 + 6m + 13)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 + 6m + 13 = 0$

$\Rightarrow (m+3)^2 - 9 + 13 = 0$

$\Rightarrow (m+3)^2 = -4$

$\Rightarrow m = -3 \pm 2\mathrm{i}$

General Solution:

$y = \mathrm{e}^{-3x}(A\cos 2x + B\sin 2x)$

27) c)

Let $y = \mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = m\mathrm{e}^{mx}$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = m^2\mathrm{e}^{mx}$

Then

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 8\frac{\mathrm{d}y}{\mathrm{d}x} + 25y = 0$

$\Rightarrow m^2\mathrm{e}^{mx} - 8m\mathrm{e}^{mx} + 25\mathrm{e}^{mx} = 0$

$\Rightarrow (m^2 - 8m + 25)\mathrm{e}^{mx} = 0$

$\mathrm{e}^{mx} \neq 0$

Auxiliary Equation:

$m^2 - 8m + 25 = 0$

$\Rightarrow (m-4)^2 - 16 + 25 = 0$

$\Rightarrow (m-4)^2 = -9$

$\Rightarrow m = 4 \pm 3\mathrm{i}$

General Solution:

$y = \mathrm{e}^{4x}(A\cos 3x + B\sin 3x)$

28) a)

$y'' - 6y' + 73y = 0$

Auxiliary Equation:

$m^2 - 6m + 73 = 0$

$\Rightarrow (m-3)^2 - 9 + 73 = 0$

$\Rightarrow (m-3)^2 = -64$

$\Rightarrow m = 3 \pm 8\mathrm{i}$

General Solution:

$y = \mathrm{e}^{3x}(A\cos 8x + B\sin 8x)$

28) b)

$16y'' - 80y' + 101y = 0$

Auxiliary Equation:

$16m^2 - 80m + 101 = 0$

$\Rightarrow 16(m^2 - 5m) + 101 = 0$

$\Rightarrow 16\left(\left(m - \frac{5}{2}\right)^2 - \frac{25}{4}\right) + 101 = 0$

$\Rightarrow 16\left(m - \frac{5}{2}\right)^2 - 100 + 101 = 0$

$\Rightarrow \left(m - \frac{5}{2}\right)^2 = -\frac{1}{16}$

$\Rightarrow m = \frac{5}{2} \pm \frac{1}{4}\mathrm{i}$

General Solution:

$y = \mathrm{e}^{\frac{5}{2}x}\left(A\cos\frac{1}{4}x + B\sin\frac{1}{4}x\right)$

29)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 6\frac{\mathrm{d}y}{\mathrm{d}x} + 10y = 0$

Auxiliary Equation:

$m^2 - 6m + 10 = 0$

$\Rightarrow (m-3)^2 - 9 + 10 = 0$

$\Rightarrow (m-3)^2 = -1$

$\Rightarrow m = 3 \pm \mathrm{i}$

General Solution:

$y = \mathrm{e}^{3x}(A\cos x + B\sin x)$

Substitute $y = 4$ and $x = 0$:

$4 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow 4 = A + 0$

$\Rightarrow A = 4$

Substitute $y = 2$ and $x = \frac{\pi}{2}$:

$2 = \mathrm{e}^{\frac{3\pi}{2}}\left(4\cos\frac{\pi}{2} + B\sin\frac{\pi}{2}\right)$

$\Rightarrow 2 = \mathrm{e}^{\frac{3\pi}{2}}(0 + B)$

$\Rightarrow B = 2\mathrm{e}^{-\frac{3\pi}{2}}$

Particular Solution:

$y = \mathrm{e}^{3x}\left(4\cos x + 2\mathrm{e}^{-\frac{3\pi}{2}}\sin x\right)$

30)

$\ddot{x} - 4\dot{x} + 13x = 0$

Auxiliary Equation:

$m^2 - 4m + 13 = 0$

$\Rightarrow (m-2)^2 - 4 + 13 = 0$

$\Rightarrow (m-2)^2 = -9$

$\Rightarrow m = 2 \pm 3\mathrm{i}$

General Solution:

$x = \mathrm{e}^{2t}(A\cos 3t + B\sin 3t)$

Substitute $x = 2$ and $t = 0$:

$2 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow 2 = A + 0$

$\Rightarrow A = 2$

$x = \mathrm{e}^{2t}(2\cos 3t + B\sin 3t)$

$\dot{x} = \mathrm{e}^{2t}(-6\sin 3t + 3B\cos 3t) + 2\mathrm{e}^{2t}(2\cos 3t + B\sin 3t)$

$= \mathrm{e}^{2t}(-6\sin 3t + 3B\cos 3t) + \mathrm{e}^{2t}(4\cos 3t + 2B\sin 3t)$

$= \mathrm{e}^{2t}\big((-6+2B)\sin 3t + (4+3B)\cos 3t\big)$

Substitute $\dot{x} = 1$ and $t = 0$:

$1 = \mathrm{e}^0\big((-6+2B)\sin 0 + (4+3B)\cos 0\big)$

$\Rightarrow 1 = 4 + 3B$

$\Rightarrow B = -1$

Particular Solution:

$x = \mathrm{e}^{2t}(2\cos 3t - \sin 3t)$

31) a)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 10\frac{\mathrm{d}y}{\mathrm{d}x} + 16y = 0$

Auxiliary Equation:

$m^2 - 10m + 16 = 0$

$\Rightarrow (m-2)(m-8)$

$\Rightarrow m = 2$ or $m = 8$

General Solution:

$y = A\mathrm{e}^{2x} + B\mathrm{e}^{8x}$

31) b)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 18\frac{\mathrm{d}y}{\mathrm{d}x} + 97y = 0$

Auxiliary Equation:

$m^2 - 18m + 97 = 0$

$\Rightarrow (m-9)^2 - 81 + 97 = 0$

$\Rightarrow (m-9)^2 = -16$

$\Rightarrow m = 9 \pm 4\mathrm{i}$

General Solution:

$y = \mathrm{e}^{9x}(A\cos 4x + B\sin 4x)$

31) c)

$\ddot{x} - 11\dot{x} - 60x = 0$

Auxiliary Equation:

$m^2 - 11m - 60 = 0$

$\Rightarrow (m+4)(m-15) = 0$

$\Rightarrow m = -4$ or $m = 15$

General Solution:

$x = A\mathrm{e}^{-4t} + B\mathrm{e}^{15t}$

31) d)

$121y'' - 110y' + 25y = 0$

Auxiliary Equation:

$121m^2 - 110m + 25 = 0$

$\Rightarrow (11m-5)^2$

$\Rightarrow m = \frac{5}{11}$

General Solution:

$y = Ax\mathrm{e}^{\frac{5}{11}x} + B\mathrm{e}^{\frac{5}{11}x}$

31) e)

$9\ddot{x} + 42\dot{x} + 49x = 0$

Auxiliary Equation:

$9m^2 + 42m + 49 = 0$

$\Rightarrow (3m + 7)^2 = 0$

$\Rightarrow m = -\frac{7}{3}$

General Solution:

$x = Ate^{-\frac{7}{3}t} + Be^{-\frac{7}{3}t}$

31) f)

$\frac{d^2y}{dx^2} - 30\frac{dy}{dx} + 225y = 0$

Auxiliary Equation:

$m^2 - 30m + 225 = 0$

$\Rightarrow (m - 15)^2$

$\Rightarrow m = 15$

General Solution:

$y = Axe^{15x} + Be^{15x}$

31) g)

$\frac{d^2y}{dx^2} + y = 0$

Auxiliary Equation:

$m^2 + 1 = 0$

$\Rightarrow m^2 = -1$

$\Rightarrow m = \pm i$

General Solution:

$y = A\cos x + B\sin x$

31) h)

$2\ddot{x} + 2\dot{x} + 5x = 0$

Auxiliary Equation:

$2m^2 + 2m + 5 = 0$

$\Rightarrow 2(m^2 + m) + 5 = 0$

$\Rightarrow 2\left(\left(m + \frac{1}{2}\right)^2 - \frac{1}{4}\right) + 5 = 0$

$\Rightarrow 2\left(m + \frac{1}{2}\right)^2 - \frac{1}{2} + 5 = 0$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 = -\frac{9}{4}$

$\Rightarrow m = -\frac{1}{2} \pm \frac{3}{2}\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-\frac{1}{2}t}\left(A\cos\frac{3}{2}t + B\sin\frac{3}{2}t\right)$

31) i)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 6\frac{\mathrm{d}y}{\mathrm{d}x} + 109y = 0$

Auxiliary Equation:

$m^2 + 6m + 109 = 0$

$\Rightarrow (m+3)^2 - 9 + 109 = 0$

$\Rightarrow (m+3)^2 = -100$

$\Rightarrow m = -3 \pm 10\mathrm{i}$

General Solution:

$y = \mathrm{e}^{-3x}(A\cos 10x + B\sin 10x)$

31) j)

$81y'' + y = 0$

Auxiliary Equation:

$81m^2 + 1 = 0$

$\Rightarrow m^2 = -\frac{1}{81}$

$\Rightarrow m = \pm\frac{1}{9}\mathrm{i}$

General Solution:

$y = A\cos\frac{1}{9}x + B\sin\frac{1}{9}x$

31) k)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 5\frac{\mathrm{d}y}{\mathrm{d}x} - 36y = 0$

Auxiliary Equation:

$m^2 + 5m - 36 = 0$

$\Rightarrow (m+9)(m-4)$

$\Rightarrow m = -9$ or $m = 4$

General Solution:

$y = A\mathrm{e}^{-9x} + B\mathrm{e}^{4x}$

31) l)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 16\frac{\mathrm{d}y}{\mathrm{d}x} + 64y = 0$

Auxiliary Equation:

$m^2 + 16m + 64 = 0$

$\Rightarrow (m+8)^2 = 0$

$\Rightarrow m = -8$

General Solution:

$y = Ax\mathrm{e}^{-8x} + B\mathrm{e}^{-8x}$

31) m)

$49\ddot{x} + 25x = 0$

Auxiliary Equation:

$49m^2 + 25 = 0$

$\Rightarrow m^2 = -\frac{25}{49}$

$\Rightarrow m = \pm\frac{5}{7}\mathrm{i}$

General Solution:

$x = A\cos\frac{5}{7}t + B\sin\frac{5}{7}t$

31) n)

$30y'' + 49y' + 20y = 0$

Auxiliary Equation:

$30m^2 + 49m + 20 = 0$

$\Rightarrow (5m + 4)(6m + 5) = 0$

$\Rightarrow m = -\frac{4}{5}$ or $m = -\frac{5}{6}$

General Solution:

$y = A\mathrm{e}^{-\frac{4}{5}x} + B\mathrm{e}^{-\frac{5}{6}x}$

31) o)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 400y = 0$

Auxiliary Equation:

$m^2 + 400 = 0$

$\Rightarrow m^2 = -400$

$\Rightarrow m = \pm 20\mathrm{i}$

General Solution:

$y = A\cos 20x + B\sin 20x$

31) p)

$\ddot{x} + 32x = 0$

Auxiliary Equation:

$m^2 + 32 = 0$

$\Rightarrow m^2 = -32$

$\Rightarrow m = \pm 4\sqrt{2}\mathrm{i}$

General Solution:

$x = A\cos\left(4\sqrt{2}t\right) + B\sin\left(4\sqrt{2}t\right)$

31) q)

$32y'' - 12y' - 35y = 0$

Auxiliary Equation:

$32m^2 - 12m - 35 = 0$

$\Rightarrow (8m + 7)(4m - 5) = 0$

$\Rightarrow m = -\frac{7}{8}$ or $m = \frac{5}{4}$

General Solution:

$y = Ae^{-\frac{7}{8}x} + Be^{\frac{5}{4}x}$

32) a)

$y = 5e^{-3x} + 6e^{4x}$

$m = -3$ or $m = 4$

Auxiliary Equation:

$(m + 3)(m - 4) = 0$

$\Rightarrow m^2 - m - 12 = 0$

Differential Equation:

$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 0$

32) b)

$y = e^{6x}(2\cos 5x + 9\sin 5x)$

$m = 6 \pm 5i$

Auxiliary Equation:

$(m - 6)^2 = -25$

$\Rightarrow m^2 - 12m + 61 = 0$

Differential Equation:

$\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 61y = 0$

32) c)

$y = 6xe^{14x} - 5e^{14x}$

$m = 14$

Auxiliary Equation:

$(m - 14)^2 = 0$

$\Rightarrow m^2 - 28m + 196 = 0$

Differential Equation:

$\frac{d^2y}{dx^2} - 28\frac{dy}{dx} + 196y = 0$

32) d)

$y = 15\cos\frac{2}{3}x + 6\sin\frac{2}{3}x$

$m = \pm\frac{2}{3}i$

Auxiliary Equation:

$m^2 = -\frac{4}{9}$

$\Rightarrow m^2 + \frac{4}{9} = 0$

Differential Equation:

$\frac{d^2y}{dx^2} + \frac{4}{9}y = 0$

33)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 0$

Auxiliary Equation:

$m^2 - m - 6 = 0$

$\Rightarrow (m+2)(m-3) = 0$

$\Rightarrow m = -2$ or $m = 3$

General Solution:

$y = A\mathrm{e}^{-2x} + B\mathrm{e}^{3x}$

Substitute $y = 10$ and $x = 0$:

$10 = A\mathrm{e}^{0} + B\mathrm{e}^{0}$

$\Rightarrow A + B = 10$ (1)

Substitute $y = 49$ and $x = \ln 2$:

$49 = A\mathrm{e}^{-2\ln 2} + B\mathrm{e}^{3\ln 2}$

$\Rightarrow 49 = A\mathrm{e}^{\ln 2^{-2}} + B\mathrm{e}^{\ln 2^{3}}$

$\Rightarrow \frac{1}{4}A + 8B = 49$ (2)

$\therefore A = 4$ and $B = 6$

Particular Solution:

$y = 4\mathrm{e}^{-2x} + 6\mathrm{e}^{3x}$

34)

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + 2y = 0$

Auxiliary Equation:

$m^2 - 2m + 2 = 0$

$\Rightarrow (m-1)^2 - 1 + 2 = 0$

$\Rightarrow (m-1)^2 = -1$

$\Rightarrow m = 1 \pm \mathrm{i}$

General Solution:

$y = \mathrm{e}^x(A\cos x + B\sin x)$

Substitute $y = 2$ and $x = 0$:

$2 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow A = 2$

$y = \mathrm{e}^x(2\cos x + B\sin x)$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^x(-2\sin x + B\cos x) + \mathrm{e}^x(2\cos x + B\sin x)$

$= \mathrm{e}^x\big((B+2)\cos x + (-2+B)\sin x\big)$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and $x = \frac{\pi}{6}$:

$0 = \mathrm{e}^{\frac{\pi}{6}}\left((B+2)\cos\frac{\pi}{6} + (-2+B)\sin\frac{\pi}{6}\right)$

$\Rightarrow 0 = (B+2)\cos\frac{\pi}{6} + (-2+B)\sin\frac{\pi}{6}$

$\Rightarrow 0 = \frac{\sqrt{3}}{2}(B+2) + \frac{1}{2}(-2+B)$

$\Rightarrow 0 = \sqrt{3}(B+2) + (-2+B)$

$\Rightarrow 0 = \sqrt{3}B + 2\sqrt{3} - 2 + B$

$\Rightarrow 2 - 2\sqrt{3} = \big(\sqrt{3}+1\big)B$

$\Rightarrow B = \frac{2-2\sqrt{3}}{\sqrt{3}+1} = -4 + 2\sqrt{3}$

Particular Solution:

$y = \mathrm{e}^x\big(2\cos x + \big(-4+2\sqrt{3}\big)\sin x\big)$